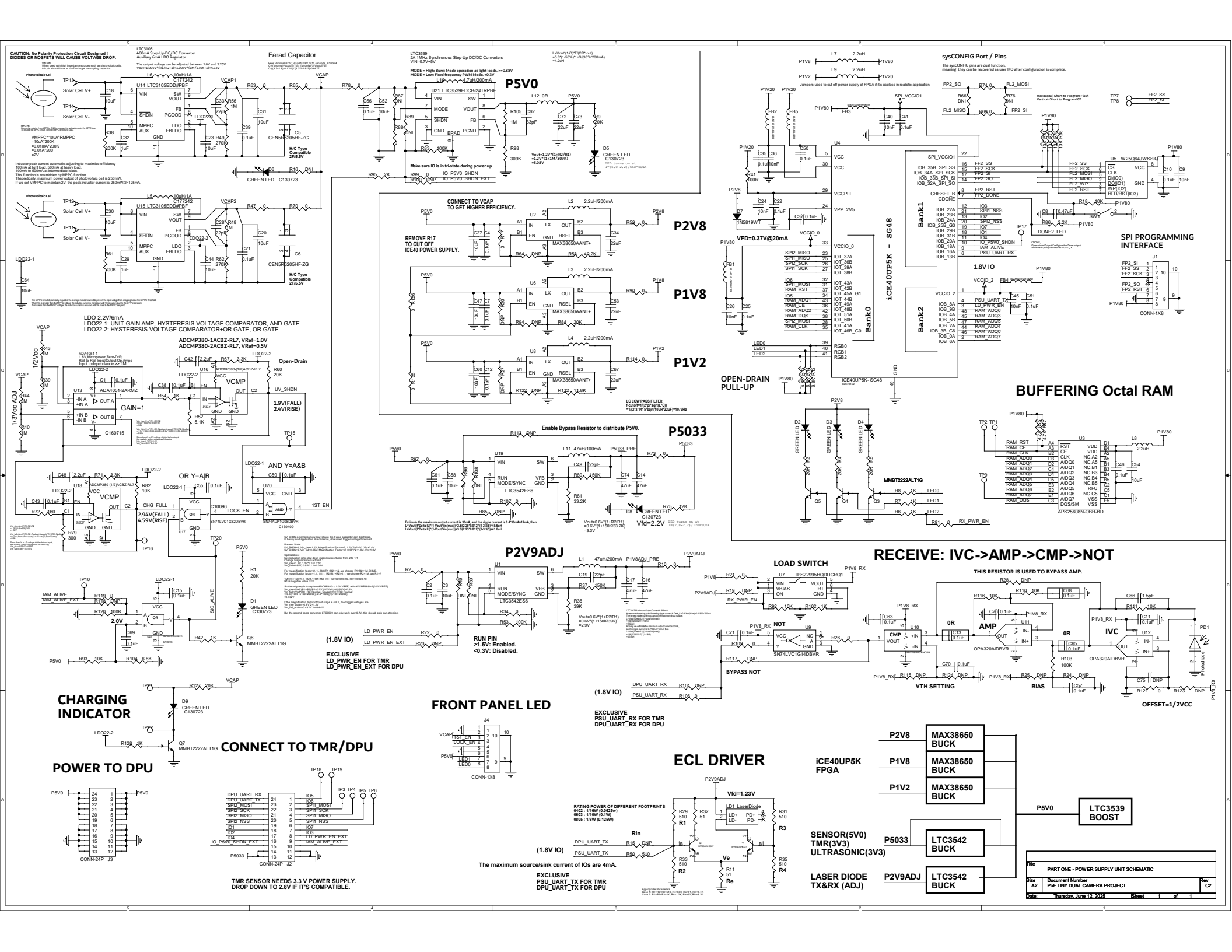

CAUTION: No Polarity Protection Circuit Designed !
DIODES OR MOSFETS WILL CAUSE VOLTAGE DROP.
Farad Capacitor
P5V0
P2V8
P1V8
P1V2
P5033
P2V9ADJ
sysCONFIG Port / Pins
SPI PROGRAMMING INTERFACE
OPEN-DRAIN PULL-UP
BUFFERING Octal RAM
RECEIVE: IVC->AMP->CMP->NOT
LOAD SWITCH
THIS RESISTOR IS USED TO BYPASS AMP.
CHARGING INDICATOR
POWER TO DPU
CONNECT TO TMR/DPU
FRONT PANEL LED
ECL DRIVER
OFFSET=1/2VCC
VTH SETTING
BIAS
BYPASS NOT
LDO 2.2V/6mA
LDO22-1: UNIT GAIN AMP, HYSTERESIS VOLTAGE COMPARATOR, AND GATE
LDO22-2: HYSTERESIS VOLTAGE COMPARATOR+OR GATE, OR GATE
ADCMP380-1ACBZ-RL7, VRef=1.0V
ADCMP380-2ACBZ-RL7, VRef=0.5V
Open-Drain
GAIN=1
OR Y=A|B
AND Y=A&B
1.9V(FALL)
2.4V(RISE)
2.9V(FALL)
4.59V(RISE)
CONNECT TO VCAP TO GET HIGHER EFFICIENCY.
REMOVE R17 TO CUT OFF ICE40 POWER SUPPLY.
Make sure IO is in tri-state during power up.
Enable Bypass Resistor to distribute P5V0.
RUN PIN
>1.5V: Enabled.
<0.3V: Disabled.
EXCLUSIVE
LD_PWR_EN FOR TMR
LD_PWR_EN_EXT FOR DPU
EXCLUSIVE
PSU_UART_RX FOR TMR
DPU_UART_RX FOR DPU
The maximum source/sink current of IOs are 4mA.
EXCLUSIVE
PSU_UART_TX FOR TMR
DPU_UART_TX FOR DPU
TMR SENSOR NEEDS 3.3 V POWER SUPPLY.
DROP DOWN TO 2.8V IF IT'S COMPATIBLE.
iCE40UP5K FPGA
P2V8 MAX38650 BUCK
P1V8 MAX38650 BUCK
P1V2 MAX38650 BUCK
P5V0 LTC3539 BOOST
SENSOR(5V0) TMR(3V3) ULTRASONIC(3V3)
P5033 LTC3542 BUCK
LASER DIODE TX&RX (ADJ)
P2V9ADJ LTC3542 BUCK
Title
PART ONE - POWER SUPPLY UNIT SCHEMATIC
Size A2
Document Number PoF TINY DUAL CAMERA PROJECT
Rev C2
Date: Thursday, June 12, 2025
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